

Ex. 22. If $2^x = 3^y = 6^{-z}$, find the value of $\left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z}\right)$.

Sol. Let $2^x = 3^y = 6^{-z} = k$. Then, $2 = k^{\frac{1}{x}}$, $3 = k^{\frac{1}{y}}$ and $6 = k^{-\frac{1}{z}}$.

$$\begin{aligned}\text{Now, } 2 \times 3 = 6 &\Leftrightarrow k^{\frac{1}{x}} \times k^{\frac{1}{y}} = k^{-\frac{1}{z}} \Leftrightarrow k^{\left(\frac{1}{x} + \frac{1}{y}\right)} = k^{-\frac{1}{z}} \\ &\Leftrightarrow \frac{1}{x} + \frac{1}{y} = -\frac{1}{z} \Leftrightarrow \frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 0.\end{aligned}$$

Ex. 23. Given $t = 2 + \sqrt[3]{4} + \sqrt[3]{2}$, determine the value of $t^3 - 6t^2 + 6t - 2$.

(M.A.T., 2005)

Sol. $t = 2 + \sqrt[3]{4} + \sqrt[3]{2} \Rightarrow t - 2 = \sqrt[3]{4} + \sqrt[3]{2} \Rightarrow (t - 2)^3 = (\sqrt[3]{4} + \sqrt[3]{2})^3$
 $\Rightarrow t^3 - 8 - 6t(t - 2) = 4 + 2 + 3 \sqrt[3]{8}(\sqrt[3]{4} + \sqrt[3]{2})$
 $\Rightarrow t^3 - 8 - 6t^2 + 12t = 6 + 3 \times 2(t - 2)$
 $\Rightarrow t^3 - 6t^2 + 12t - 8 = 6 + 6t - 12 \Rightarrow t^3 - 6t^2 + 6t - 2 = 0.$

EXERCISE

(OBJECTIVE TYPE QUESTIONS)

Directions: Mark (✓) against the correct answer:

1. $\sqrt[3]{5}$ is a surd of the order (R.R.B., 2008)

- (a) $\frac{1}{3}$ (b) 1
(c) 2 (d) 3

2. $5^0 \times 8 = ?$ (R.R.B., 2006)

- (a) 0 (b) 8
(c) 40 (d) 200

3. Which of the following are equal in value?

- I. 4^1 II. 1^4
III. 4^0 IV. 0^4
(a) I and II (b) II and III
(c) III and IV (d) I and IV

4. If $289 = 17^{\frac{1}{5}x}$, then $x = ?$ (Bank P.O., 2009)

- (a) $\frac{2}{5}$ (b) 8
(c) 16 (d) 32
(e) None of these

5. $(81)^4 \div (9)^5 = ?$ (Agriculture Officers', 2009)

- (a) 9 (b) 81
(c) 729 (d) 6561
(e) None of these

6. The value of $\left(\frac{9^2 \times 18^4}{3^{16}}\right)$ is (R.R.B., 2006)

- (a) $\frac{3}{2}$ (b) $\frac{4}{9}$
(c) $\frac{16}{81}$ (d) $\frac{32}{243}$

7. $[4^3 \times 5^4] \div 4^5 = ?$

(Bank Recruitment, 2008)

- (a) 29.0825 (b) 30.0925
(c) 35.6015 (d) 39.0625
(e) None of these

8. $9^3 \times 6^2 \div 3^3 = ?$

(I.C.A.D.O., 2007)

- (a) 948 (b) 972
(c) 984 (d) 1012
(e) None of these

9. $(19)^{12} \times (19)^8 \div (19)^4 = (19)^?$ (Bank Recruitment, 2008)

- (a) 6 (b) 8
(c) 12 (d) 24
(e) None of these

10. $(64)^4 \div (8)^5 = ?$

(Agriculture Officer's, 2008)

- (a) $(8)^8$ (b) $(8)^2$
(c) $(8)^{12}$ (d) $(8)^4$
(e) None of these

11. $(1000)^{12} \div (10)^{30} = ?$

(Bank P.O., 2008)

- (a) $(1000)^2$ (b) 10
(c) 100 (d) $(100)^2$
(e) None of these

12. $(3)^8 \times (3)^4 = ?$

(Bank P.O., 2009)

- (a) $(27)^3$ (b) $(27)^5$
(c) $(729)^2$ (d) $(729)^3$
(e) None of these

13. $\frac{343 \times 49}{216 \times 16 \times 81} = ?$

(Bank P.O., 2010)

- (a) $\frac{7^5}{6^7}$ (b) $\frac{7^5}{6^8}$
(c) $\frac{7^6}{6^7}$ (d) $\frac{7^4}{6^8}$
(e) None of these